representation using external lexicons for identifying terms that can be potential indicators of subjectivity. We can use the count of terms present in external subjectivity and emotion lexicons as an additional feature in our feature vector. The constructed feature vector for the input is then fed to a standard supervised learning based classifier.

We can also experiment with a baseline neural network classifier. The input is a question-embedding representation, which can be as simple as the concatenation of the word vectors of the words present in the question. This question-embedding can be passed through an LSTM (long short-term memory) encoder-decoder to create an encoded representation that can then be passed through a feed forward neural network, followed by a softmax binary classifier. As in the case of standard machine learning based classifiers, the encoded embedding input to the feed forward neural network classifier can be augmented with additional features, as we discussed earlier.

However, note that since each category of the insincere question is extremely different from others, such a general text classifier may not perform well, since it may not be able to delineate the different categories well. We can also explore multiple classifiers – one for each of the above four categories of insincere questions and combine the outputs into a voting classifier to decide the overall label. I would like to suggest that our readers experiment both with a simple baseline classifier as well as with a multiple classifiers approach. We will discuss more about the multiple classifiers approach in next month's column.

If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month and a very happy new year! END

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